

Time Series Regression

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Basic time series concepts

Additive decomposition models

Multiplicative decomposition models

A warning about trend-based forecasting

Autoregressive models

Special kinds of $AR(1)$ models

Making predictions with AR models

Combining decomposition and AR models

The market model



What are time series?

- Data where the cases represent time: data collected every day, month, year, etc.
- Time series are important for both **explaining** how variables change over time and **forecasting** the future
- Examples of time series data:
 - Google's closing daily stock price every day in 2020
 - Inventory levels of each item at a retail store at the end of every week in 2020
 - Number of new COVID cases in the US each day since the start of the pandemic
 - Apple's quarterly revenue since 2009



Anatomy of a time series

Some notation:

t = time

Y_t = the value of the variable we are interested in, at time t

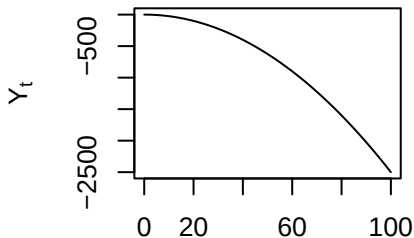
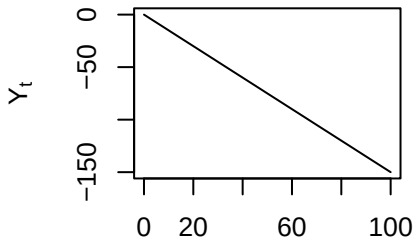
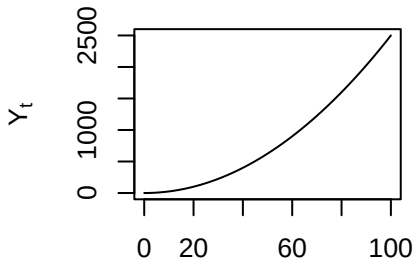
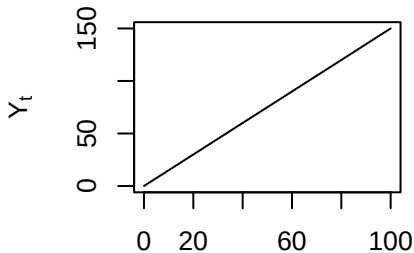
Y_t may be composed of one or more components:

- Trend
- Seasonal
- Cyclical
- Random



Trend component

An **trend** is persistent upwards or downwards movement in the data





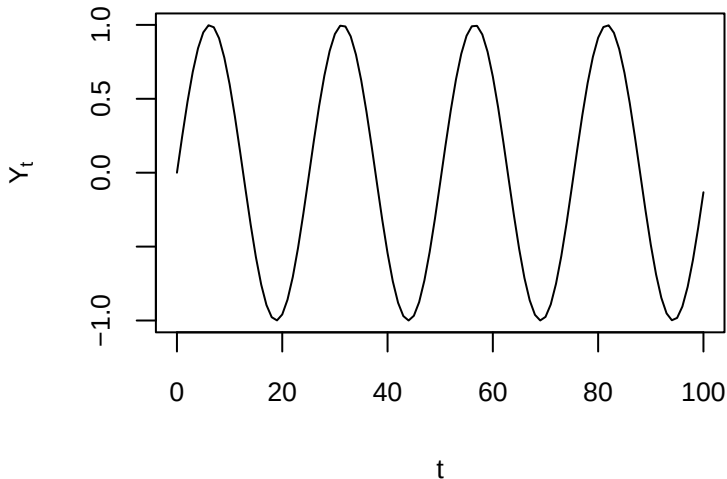
Trend component

- The trend doesn't need to be linear!
- Example: Moore's Law (accelerating increase of transistor count)
- Example: US population
- A time series with no trend is **stationary**



Seasonal component

Seasonal fluctuation occurs when predictable up or down movements occur over a regular interval.





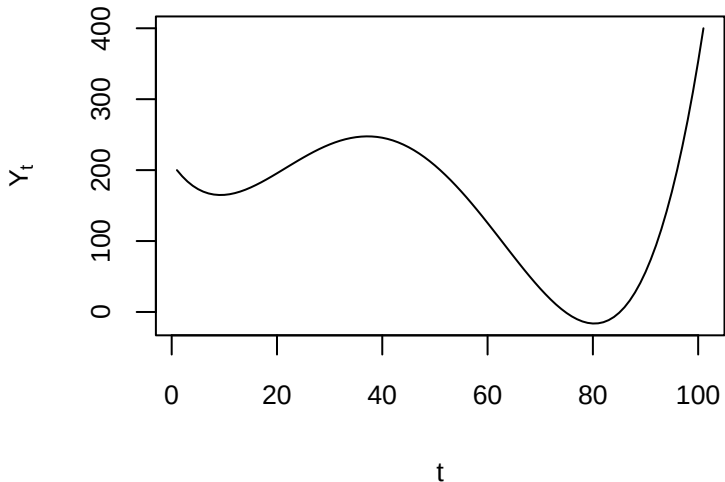
Seasonal component

- The ups and downs must occur over a regular interval (e.g., every month, or every year)
- Example: Highway traffic volume is highest during rush hour every day
- Example: Supermarket sales may be highest every month right after common paydays like the 15th and 30th



Cyclic component

Cyclic fluctuations occur at unpredictable intervals, e.g. due to changing business or economic conditions.





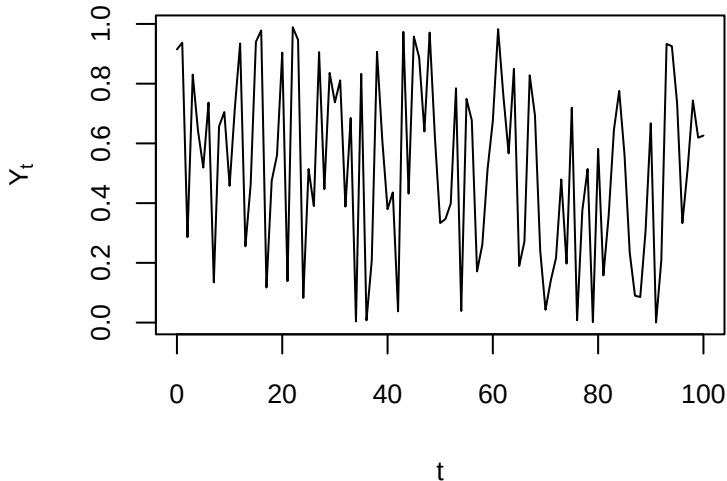
Cyclic component

- In contrast to seasonal fluctuations, cyclic fluctuations do not occur at regular, predictable intervals
- It may be possible to predict cyclic components based on some other (non-time) variable
- Example: Restaurant sales dropped dramatically in 2020 due to COVID, as people ate out less
- Example: Sales of bell bottoms rose in the 60s and 70s, declined by the 80s, and then had a resurgence in the 90s



Remainder/Error component

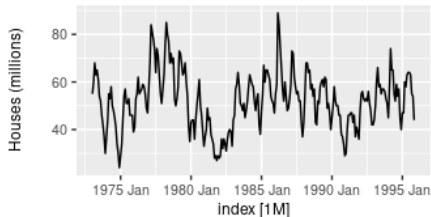
Any real time series will always have **random noise** as well, which can't be predicted or forecast.



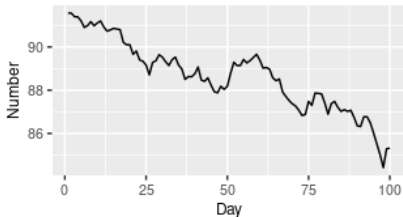


Which component(s) do you see?

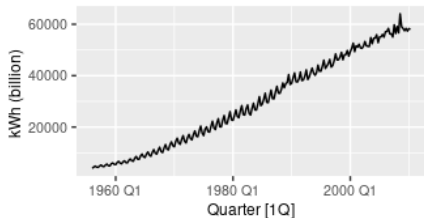
Sales of new one-family houses, USA



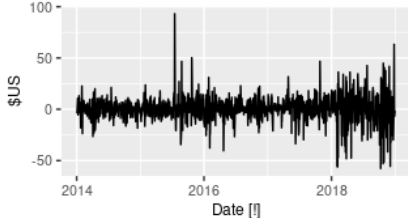
US treasury bill contracts



Australian quarterly electricity production



Daily changes in Google closing stock price





Putting these together

Real time series will usually include a combination of these four components. We will model the time series Y_t either additively:

$$Y_t = \text{trend} + \text{seasonal} + \text{random} = T_t + S_t + E_t$$

Or multiplicatively:

$$Y_t = \text{trend} \cdot \text{seasonal} \cdot \text{random} = T_t \cdot S_t \cdot E_t$$

(E_t consists of both the cyclic and error components, as both are unpredictable.)

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Additive models

- All components (T_t, S_t, E_t) are in the units of Y
- Most appropriate when seasonal fluctuations are consistent (do not increase or decrease over time)
- The trend component T_t is a function of t (e.g., linear or quadratic)
- The seasonal component S_t is a set of dummy variables representing “seasons”
- So we can estimate additive models using regular old regression!

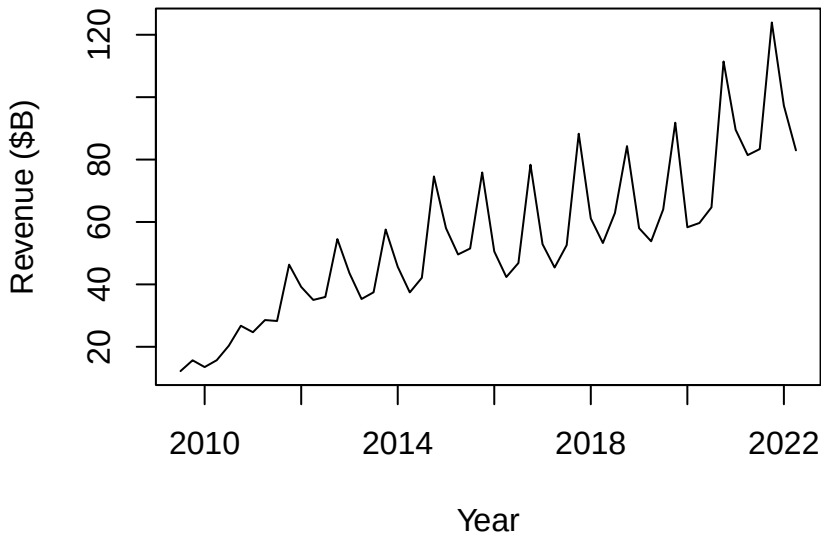


Additive decomposition

1. Run a regression predicting Y as a function of:
 - $t, t^2, \log(t)$ etc (the **trend component** T_t)
 - Dummy variables for the seasons (the **seasonal component** S_t)
2. To make a prediction for Y , plug into the model!
3. The residuals of this model correspond to the **error component** E_t



Apple quarterly revenue



What components do you see here?



Apple data

```
head(apple)
```

	Period	Time	Year	Quarter	Revenue
1	1	2009.50	2009	Q3	12.21
2	2	2009.75	2009	Q4	15.68
3	3	2010.00	2010	Q1	13.50
4	4	2010.25	2010	Q2	15.70
5	5	2010.50	2010	Q3	20.34
6	6	2010.75	2010	Q4	26.74

(Note that the fractional part of year indicates the quarter.)

```
linear.trend <- lm(Revenue ~ Period, data=apple)
summary(linear.trend)
```

Call:

```
lm(formula = Revenue ~ Period, data = apple)
```

Residuals:

Min	1Q	Median	3Q	Max
-19.88	-8.13	-3.24	6.74	36.20

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.452	3.758	4.64	2.5e-05 ***
Period	1.406	0.123	11.39	1.7e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

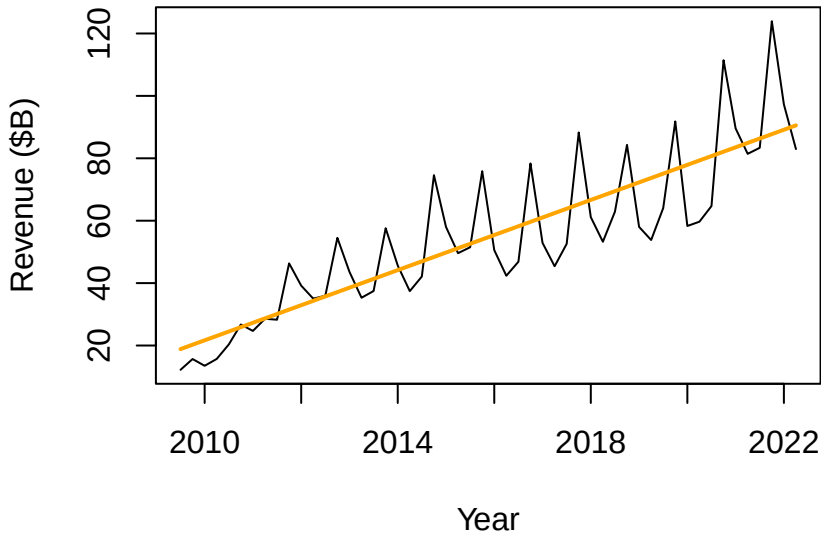
Residual standard error: 13.4 on 50 degrees of freedom

Multiple R-squared: 0.722, Adjusted R-squared: 0.716

F-statistic: 130 on 1 and 50 DF, p-value: 1.67e-15



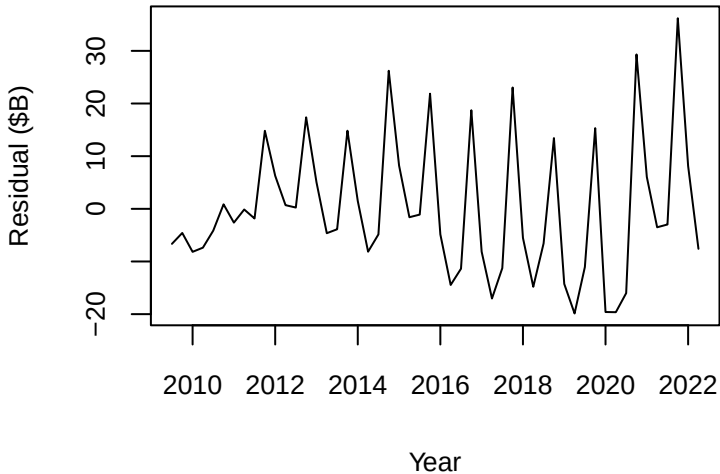
Fitting a linear trend component





Detrended data

The residuals from this model show the “detrended” data (but there’s still some trend left, and clear seasonality that we are ignoring!):





Adding seasonality

Let's incorporate seasonality into our model, by adding dummies for the quarter (leaving out one, like any categorical variable):

```
linear.trend.seasonality <- lm(Revenue ~ Period + Quarter, data=apple)
summary(linear.trend.seasonality)
```

Call:

```
lm(formula = Revenue ~ Period + Quarter, data = apple)
```

Residuals:

Min	1Q	Median	3Q	Max
-21.70	-5.13	1.09	5.30	18.33

Coefficients:

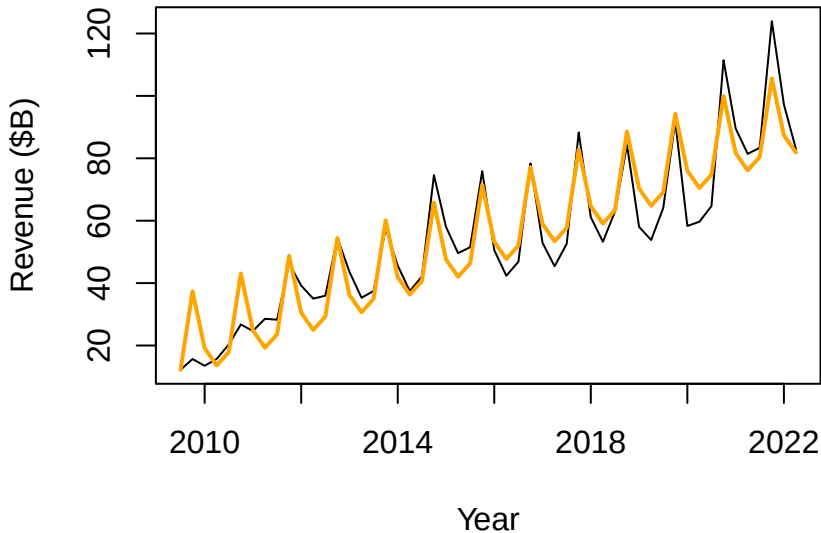
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.8822	3.1273	4.76	0.00001892 ***
Period	1.4215	0.0776	18.32	< 2e-16 ***
QuarterQ2	-6.9492	3.2850	-2.12	0.04 *
QuarterQ3	-4.0878	3.2878	-1.24	0.22
QuarterQ4	19.6569	3.2850	5.98	0.00000028 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.37 on 47 degrees of freedom



Linear trend with seasonality





Interpreting the coefficients

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	14.8822	3.1273	4.76	0.00001892	***
Period	1.4215	0.0776	18.32	< 2e-16	***
QuarterQ2	-6.9492	3.2850	-2.12	0.04	*
QuarterQ3	-4.0878	3.2878	-1.24	0.22	
QuarterQ4	19.6569	3.2850	5.98	0.00000028	***

- Q2's are expected to be \$6.9B worse than Q1's
- Q3's are expected to be \$4.1B worse than Q1's
- Q4's are expected to be \$19.7B better than Q1's



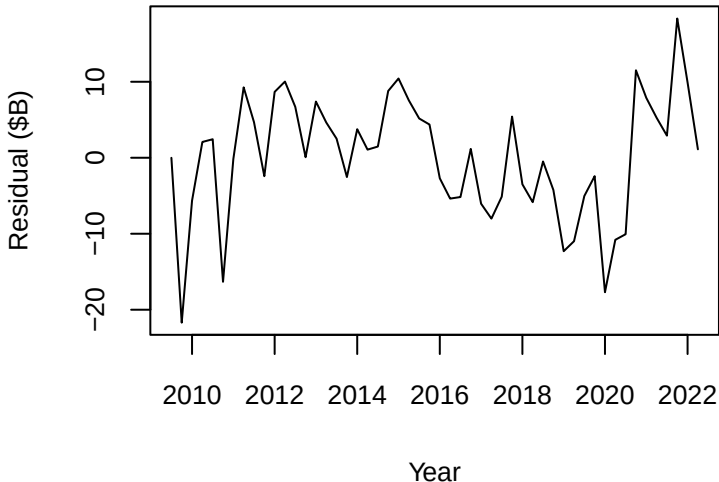
What does our model predict for the future?

- What does the final model predict for Apple in 2023 Q1?
- What does the final model predict for Apple in 2030 Q1? (Should we trust that prediction?)



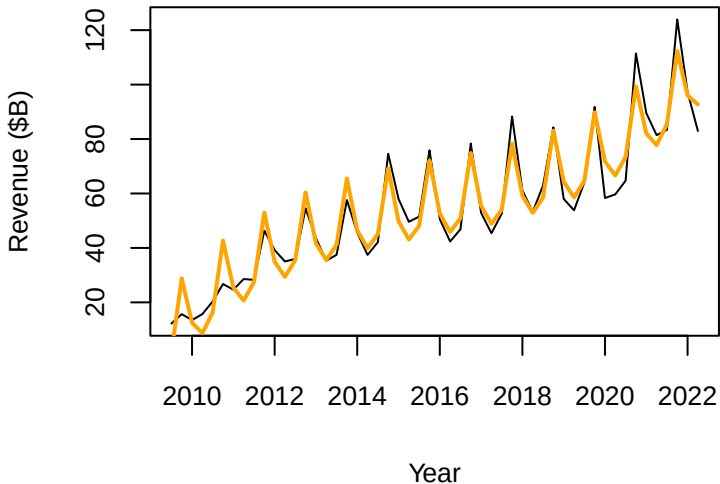
Detrended and seasonalized data

The residuals from this model show the “detrended and deasonalized” data (but there's still some trend left!):





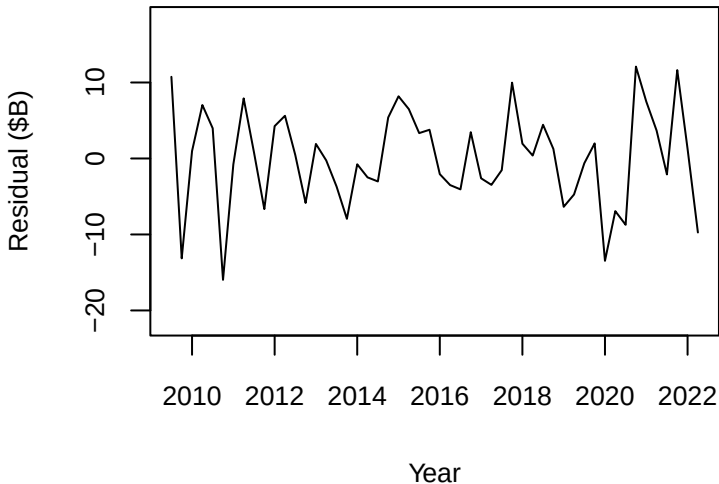
Improving the model by fitting a cubic trend





Detrended and deasonalized data (cubic model)

The residuals from this model show the “detrended and deasonalized” data—much better! What’s left looks random:



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- Only the trend component T_t is in the units of Y
- The seasonal and error components are multipliers (e.g., applying a percentage rather than absolute change to the reference season)
- We can estimate this using regression by taking logs of both sides of the equation $Y_t = T_t \cdot S_t \cdot E_t$:

$$\log(Y_t) = \log(T_t) + \log(S_t) + \log(E_t)$$

- A multiplicative model is appropriate when the magnitude of the seasonality increases (or decreases) over time

```
log.trend.seasonality <- lm(log(Revenue) ~ log(Period) + Quarter, data=apple)
summary(log.trend.seasonality)
```

Call:

```
lm(formula = log(Revenue) ~ log(Period) + Quarter, data = apple)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.2083	-0.0810	-0.0236	0.0954	0.3540

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.1936	0.0719	30.53	< 2e-16 ***
log(Period)	0.5493	0.0204	26.95	< 2e-16 ***
QuarterQ2	-0.1248	0.0507	-2.46	0.017 *
QuarterQ3	-0.0453	0.0508	-0.89	0.377
QuarterQ4	0.3166	0.0507	6.25	0.00000011 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

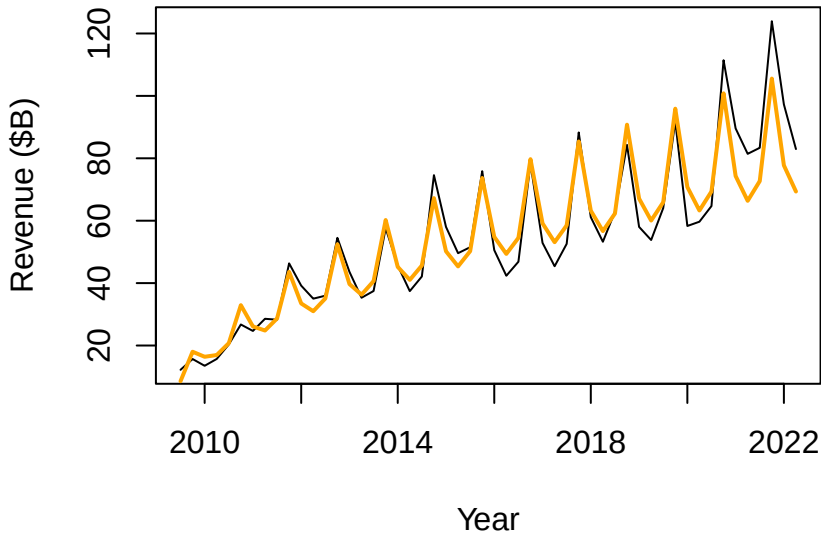
Residual standard error: 0.129 on 47 degrees of freedom

Multiple R-squared: 0.945, Adjusted R-squared: 0.94

F-statistic: 202 on 4 and 47 DF, p-value: <2e-16



Multiplicative model captures increasing variation





Writing out a multiplicative model

Our model of the trend component is:

$$\log(\widehat{\text{Rev}}) = 2.19 + 0.55 \log(t) - 0.12(\text{Q2}) - 0.05(\text{Q3}) + 0.32(\text{Q4})$$



Interpreting multiplicative model coefficients

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.1936	0.0719	30.53	< 2e-16	***
log(Period)	0.5493	0.0204	26.95	< 2e-16	***
QuarterQ2	-0.1248	0.0507	-2.46	0.017	*
QuarterQ3	-0.0453	0.0508	-0.89	0.377	
QuarterQ4	0.3166	0.0507	6.25	0.00000011	***

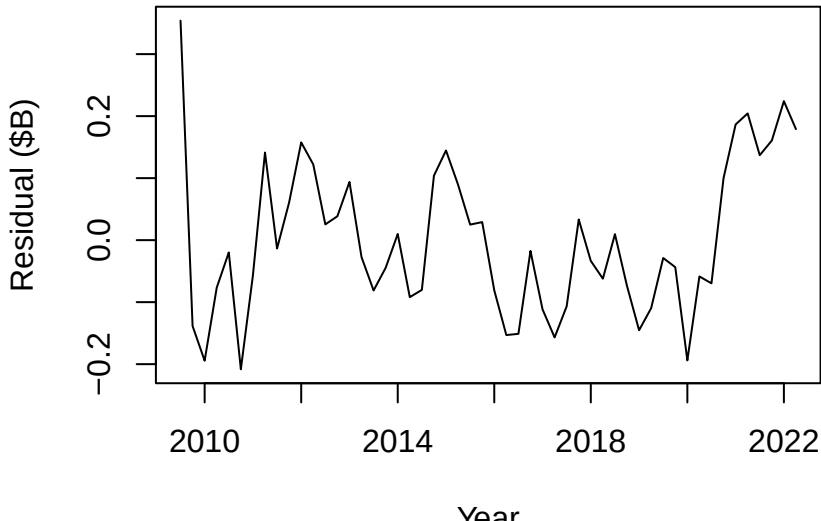
Because this is a model where we are predicting $\log(Y)$:

- Q2's are expected to be 12% below Q1's
- Q3's are expected to be 5% below Q1's
- Q4's are expected to be 32% above Q1's



Detrended/deseasonalized (multiplicative)

The residuals from this model show the “detrended and deseasonalized” data—good! What’s left looks (pretty) random:



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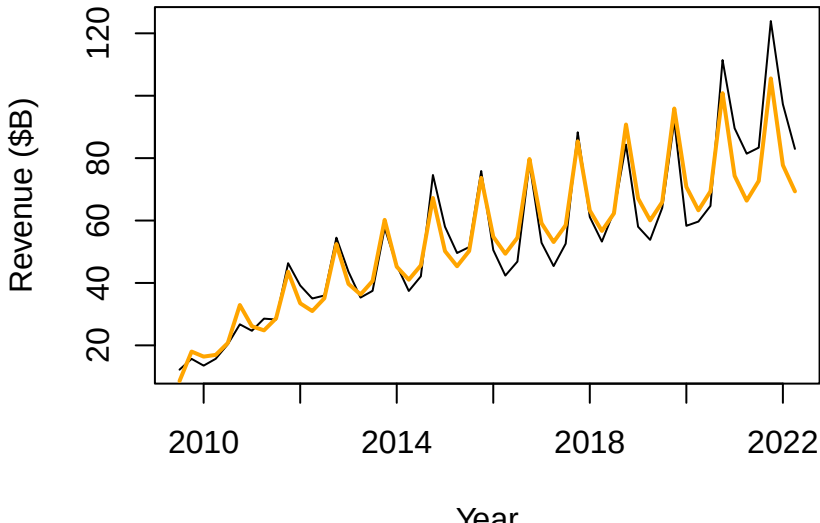
The market model

Sometimes forecasting using a trend-based model is dangerous—things change!



Have things changed?

Apple seems to be consistently outperforming the model in the last couple of years.



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Limitations of decomposition models

Previously, we created **decomposition** models that predict Y_t as a function of time:

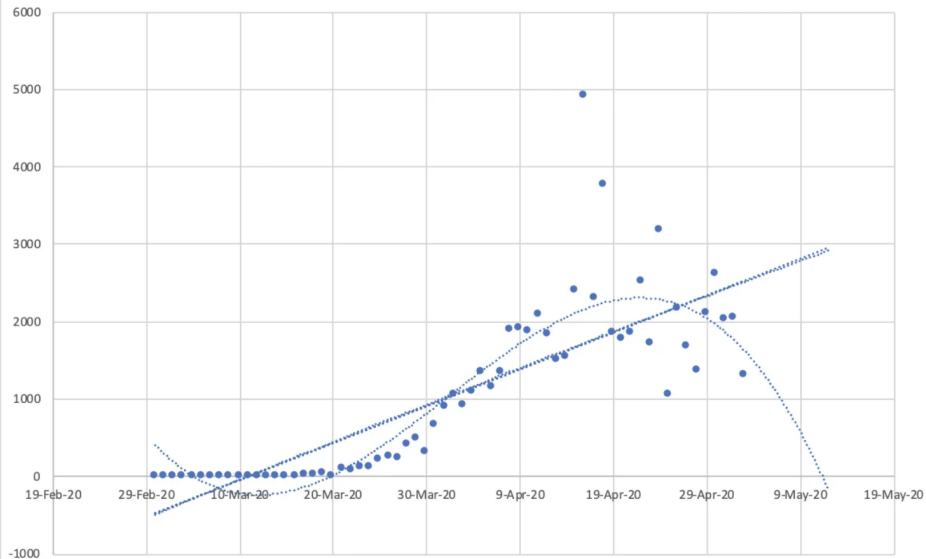
$$Y_t = \text{trend} + \text{seasonal} + \text{error} = T_t + S_t + E_t$$

$$Y_t = \text{trend} \times \text{seasonal} \times \text{error} = T_t \times S_t \times E_t$$

But sometimes forecasting using a trend-based model is dangerous or unworkable!

We might also be leaving some predictive ability on the table if E_t includes a cyclical component.

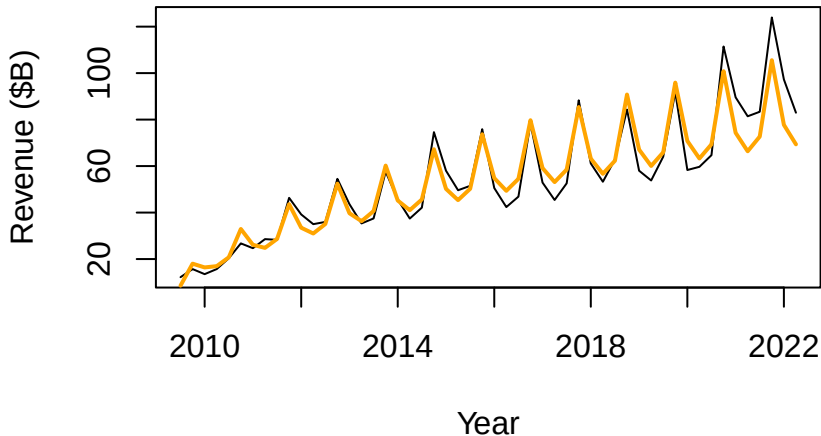
Covid Death Scatterplot With Linear and Cubic Trendlines





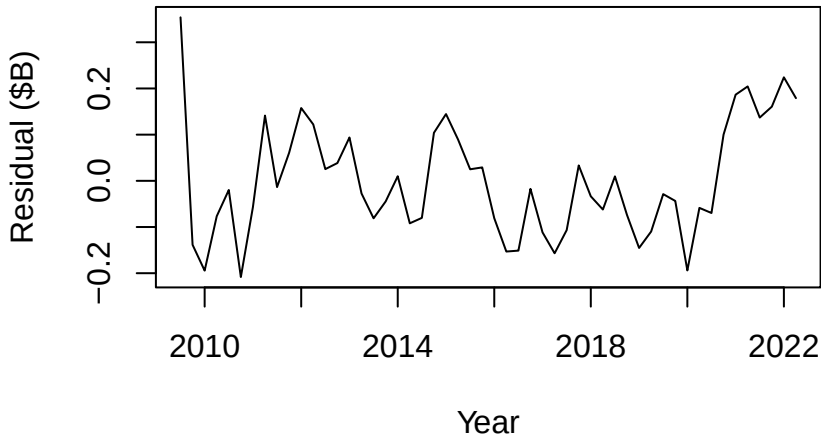
Recall: Multiplicative model for Apple's revenue

```
lm(log(Revenue) ~ log(Period) + Quarter, data=apple)
```





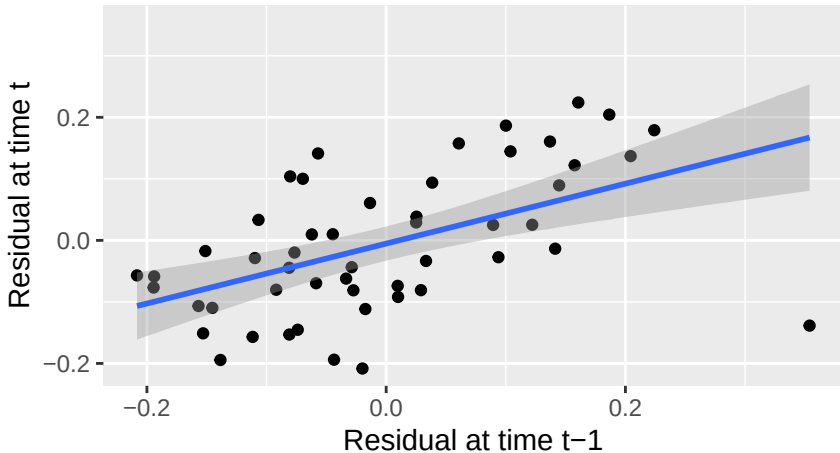
Recall: Multiplicative model for Apple's revenue



We thought our residuals looked pretty good, but...



Past prediction errors predict future errors

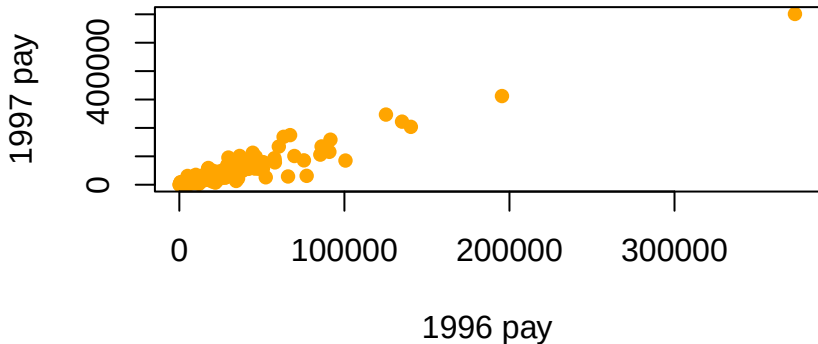


How can we use this to improve our model?

What factors do you think would be the best predictors of how much a company pays their CEO in a given year?

What factors do you think would be the best predictors of how much a company pays their CEO in a given year?

If we want to predict CEO pay in 1997, knowing the CEO's pay in 1996 seems to help a lot:



A model like this incorporates a lot of other variables that might not be available in our data set!



Autoregressive models

Key idea: Predict Y_t as a function of Y_{t-1} :

$$\hat{Y}_t = \hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$$

- Y_{t-1} is called the “1st lag” of Y (e.g., if $t = 1997$, then $Y_t = \text{pay1997}$ and $Y_{t-1} = \text{pay1996}$)



Autoregressive models

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$$\hat{Y}_t = \hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$$

- Y_{t-1} is called the “1st lag” of Y (e.g., if $t = 1997$, then $Y_t = \text{pay1997}$ and $Y_{t-1} = \text{pay1996}$)
- This is called **autoregressive** because it predicts the values of a time series based on previous values



Autoregressive models

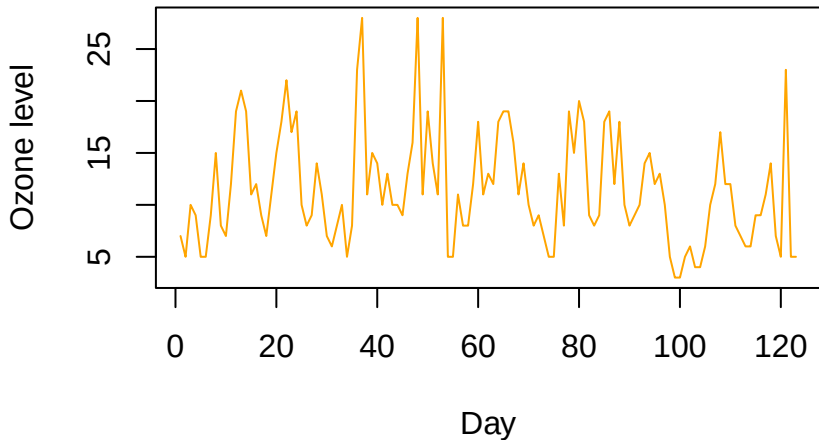
Key idea: Predict Y_t as a function of Y_{t-1} :

$$\hat{Y}_t = \hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$$

- Y_{t-1} is called the “1st lag” of Y (e.g., if $t = 1997$, then $Y_t = \text{pay1997}$ and $Y_{t-1} = \text{pay1996}$)
- This is called **autoregressive** because it predicts the values of a time series based on previous values
- This is an AR(1) model (an AR(k) model would predict 1997 pay from the previous k years)



AR(1) example: Daily ozone in Houston



Decomposing Y_t into trend and seasonal components probably isn't going to work—no clear trend or seasonality!

```
summary(lm(ozone ~ day, data=ozone))
```

Call:

```
lm(formula = ozone ~ day, data = ozone)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.97	-4.04	-1.09	2.79	16.27

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.0152	0.9772	13.32	<2e-16 ***
day	-0.0242	0.0137	-1.77	0.079 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

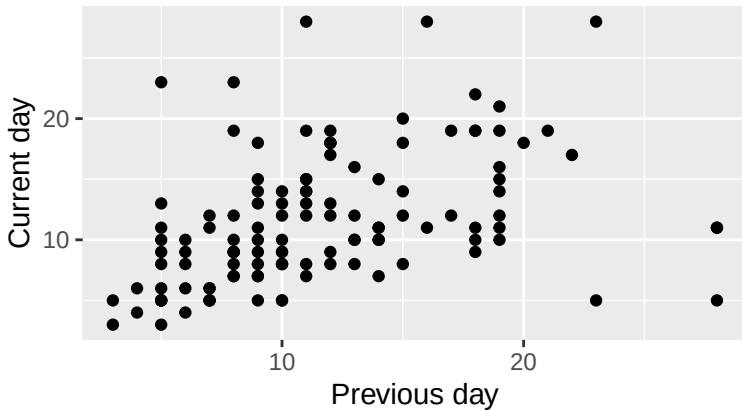
Residual standard error: 5.4 on 121 degrees of freedom

Multiple R-squared: 0.0253, Adjusted R-squared: 0.0173

F-statistic: 3.14 on 1 and 121 DF, p-value: 0.0788

Can we use yesterday's ozone level to predict today's?

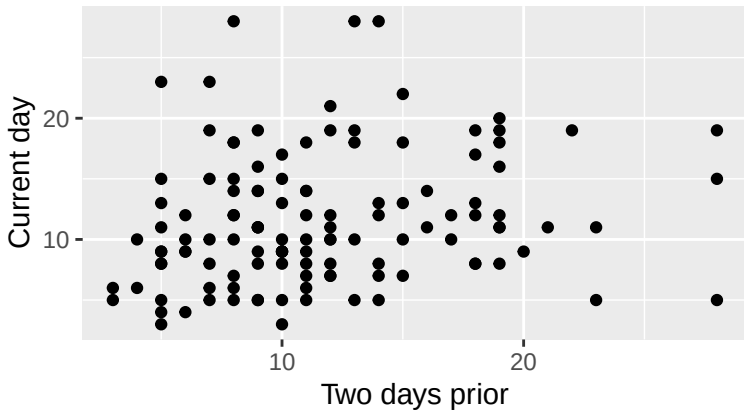
Ozone level



$$r = 0.4$$

How about ozone from two days ago?

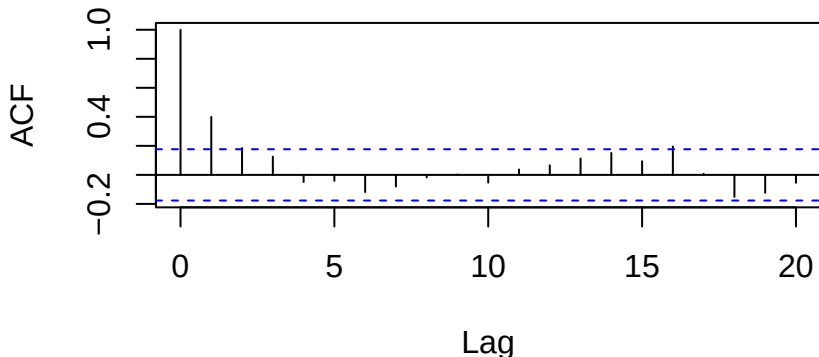
Ozone level



$r = 0.19$ (But can you think of any reason this correlation might be misleading?)

Autocorrelation is the correlation of Y_t with its lags (Y_{t-1} , Y_{t-2} , etc). The autocorrelation function (ACF) displays these:

```
acf(ozone$ozone)
```



Autocorrelations outside of the dashed blue lines are statistically significant.

```
ozone <- ozone %>% mutate(lag1=lag(ozone), lag2=lag(ozone, 2))
ozone.model <- lm(ozone ~ lag1, data=ozone)
summary(ozone.model)
```

Call:

```
lm(formula = ozone ~ lag1, data = ozone)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.19	-3.46	-1.11	2.68	16.68

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.8745	1.0698	6.43	0.0000000028 ***
lag1	0.4042	0.0838	4.82	0.0000041974 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5 on 120 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.162, Adjusted R-squared: 0.155

F-statistic: 23.3 on 1 and 120 DF, p-value: 0.0000042

```
ozone <- ozone %>% mutate(lag1=lag(ozone), lag2=lag(ozone, 2))
ozone.model2 <- lm(ozone ~ lag1 + lag2, data=ozone)
summary(ozone.model2)
```

Call:

```
lm(formula = ozone ~ lag1 + lag2, data = ozone)
```

Residuals:

Min	1Q	Median	3Q	Max
-12.87	-3.35	-1.12	2.73	16.54

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.7241	1.2604	5.33	0.00000047 ***
lag1	0.3835	0.0919	4.17	0.00005763 ***
lag2	0.0370	0.0922	0.40	0.69

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5 on 118 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.16, Adjusted R-squared: 0.145

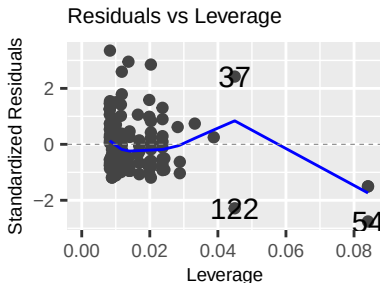
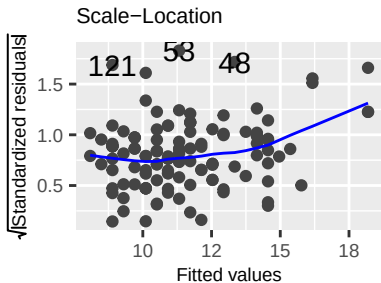
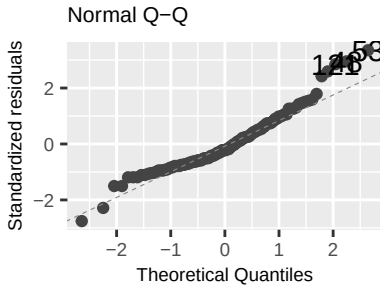
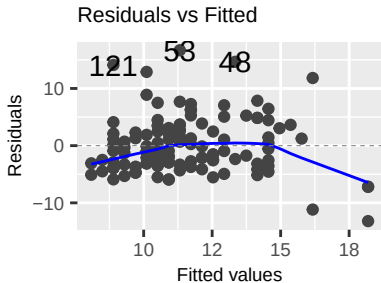
F-statistic: 11.2 on 2 and 118 DF, p-value: 0.000035



Checking assumptions of an AR(1) model

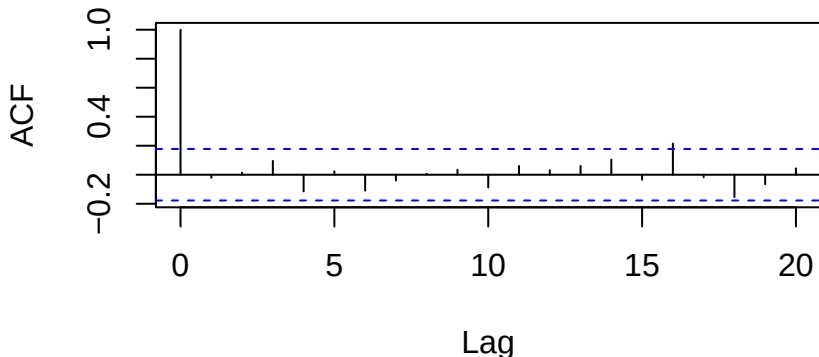
- Linearity, Homoscedasticity, Normality: Check using residual plot (linearity + homoscedasticity), Q-Q plot (normality), scale/location (homoscedasticity) like any other regression model
- Independence: Since this is a time series, we can actually check this by looking at the autocorrelation of the *residuals* (we want *no* significant autocorrelation)

Checking assumptions of an AR(1) model



Checking independence in an AR(1) model

```
acf(residuals(ozone.model))
```



We expect 5% of autocorrelations to be significant just by chance at $\alpha = 0.05$, so this ACF suggests we are OK on independence!

Basic time series concepts

Additive decomposition models

Multiplicative decomposition models

A warning about trend-based forecasting

Autoregressive models

Special kinds of AR(1) models

Making predictions with AR models

Combining decomposition and AR models

The market model



Special kinds of AR(1) models

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t$$

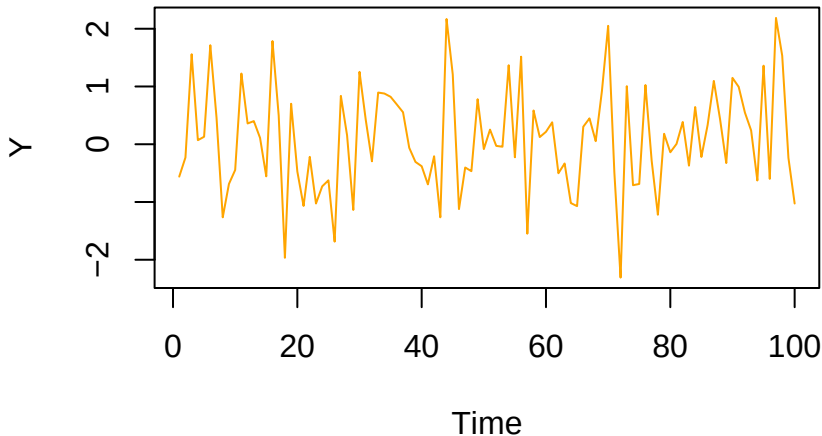
- If $\beta_0 = 0$ and $\beta_1 = 0$, then this is **white noise**
- If $\beta_0 \neq 0$ and $\beta_1 = 0$, then this is a **random sample**
- If $\beta_0 = 0$ and $\beta_1 = 1$, then this is a **random walk**
- If $\beta_0 \neq 0$ and $\beta_1 = 1$, then this is a **random walk with drift**
- If $\beta_1 < 1$, then Y_t will oscillate around the mean



White noise

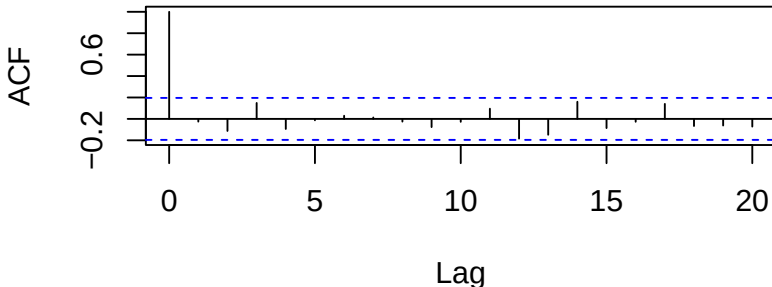
A white noise time series is one where there is no pattern at all—each value is totally unpredictable.

$$Y_t = \epsilon_t$$



Could my time series be white noise? If:

1. The ACF shows no significant autocorrelation:



2. And we fit an AR(1) model and find that that 0 falls within the 95% CIs for both β_0 and β_1 :

	2.5 %	97.5 %
(Intercept)	-0.0849	0.284
lag.y	-0.2286	0.176

Then there is no statistically significant evidence that the series isn't white noise



A random sample time series is one where each value is just independently sampled from the same distribution:

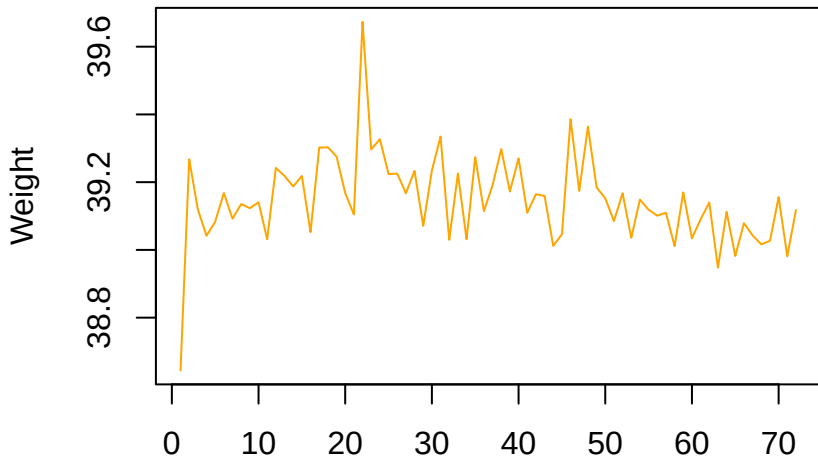
$$Y_t = \beta_0 + \epsilon_t$$

- There is no Y_{t-1} term: like with white noise, it doesn't matter what the previous value was
- β_0 represents the mean of the distribution that the values are being sampled from



Random sample

In the Folgers factory, 39-oz coffee cans are filled up with slightly different amounts of coffee each hour (just due to random variation with the machine):





Could it be a random sample?

1. The ACF shows no significant autocorrelation
2. We fit an AR(1) model and find that 0 falls within the 95% CI for β_1 but *not* for β_0 :

	2.5 %	97.5 %
(Intercept)	27.482	44.112
lag.wt	-0.127	0.298

Then:

- There is **statistically significant evidence** that the series **isn't** white noise
- There is **no statistically significant evidence** that the series **isn't** a random sample



A random walk time series is one where each value is predicted to be the previous value, plus some drift (maybe):

$$Y_t = \beta_0 + Y_{t-1} + \epsilon_t$$

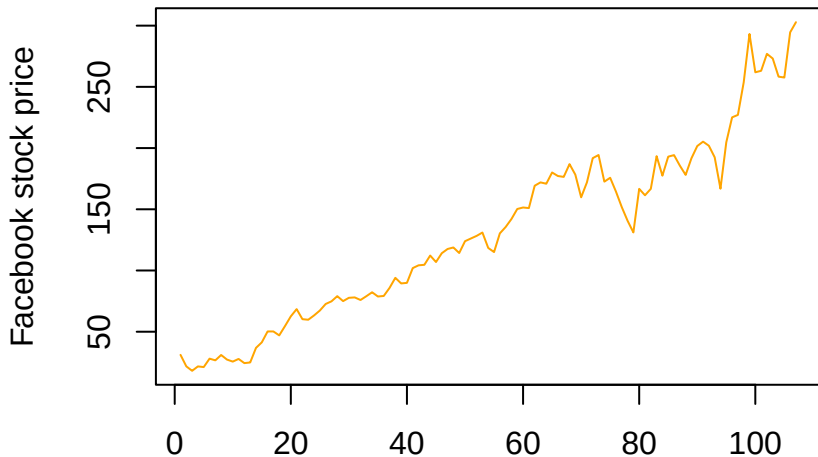
- β_0 is the “drift” (could be 0) and is in some sense the overall trend
- A time series is a random walk if the *changes* are a random sample time series:

$$Y_t - Y_{t-1} = \beta_0 + \epsilon_t$$



Facebook stock price

Random walks are often a good choice for stock prices, like the Facebook stock price over a 107-month period from June 2012 to last year:





Could it be a random walk?

We fit an AR(1) model and find that 1 falls within the 95% CI for β_1 (and zero doesn't):

	2.5 %	97.5 %
(Intercept)	-2.766	6.64
lagFB	0.973	1.04

Then:

- There is statistically significant evidence that the series isn't white noise or a random sample
- There is no statistically significant evidence that the series isn't a random walk

Bonus: If this series isn't truly a random walk, do you think it might behave similarly to a random walk anyway? Why?

Basic time series concepts

Additive decomposition models

Multiplicative decomposition models

A warning about trend-based forecasting

Autoregressive models

Special kinds of $AR(1)$ models

Making predictions with AR models

Combining decomposition and AR models

The market model



Making predictions in time series

Type	Model	Predicted Y_t
White noise	$Y_t = \epsilon_t$	0
Random sample	$Y_t = \beta_0 + \epsilon_t$	$\hat{\beta}_0$ (or average Y)
Random walk	$Y_t = \beta_0 + Y_{t-1} + \epsilon_t$	$\hat{\beta}_0 + Y_{t-1}$
General AR(1)	$Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t$	$\hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$

Basic time series concepts

Additive decomposition models

Multiplicative decomposition models

A warning about trend-based forecasting

Autoregressive models

Special kinds of $AR(1)$ models

Making predictions with AR models

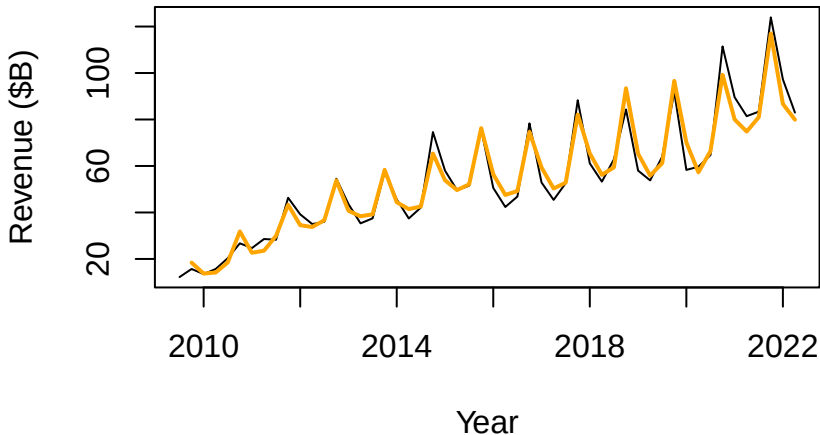
Combining decomposition and AR models

The market model



Combining decomposition and AR models

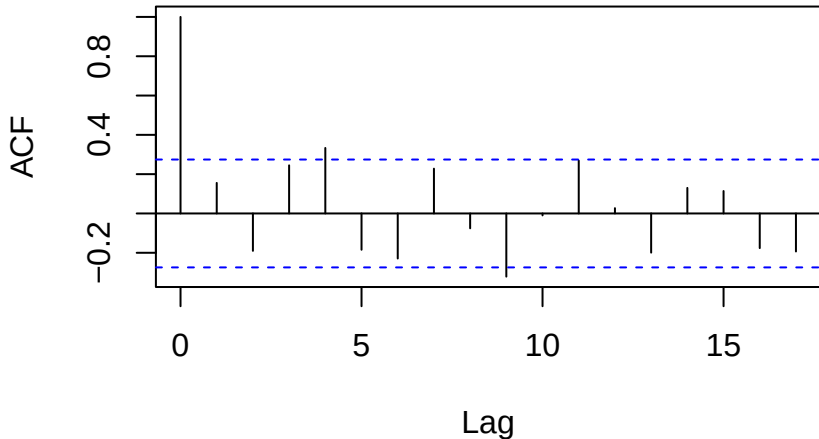
```
apple = apple %>% mutate(lag1 = lag(Revenue))  
apple_model <- lm(log(Revenue) ~ log(Period) + Quarter  
                  + log(lag1), data=apple)
```





One lag fixes residual autocorrelation

```
acf(residuals(apple_model))
```





Why lags and not residuals?

- We started with the observation that the residuals (prediction errors) were autocorrelated. So why not use lagged *residuals* as predictors?
- We need a model to compute residuals, so the model can't depend on the residuals!
- Y_{t-1} (in this case $\log(Y_{t-1})$) contains all the relevant predictive information to eliminate lag-1 autocorrelation in the residuals



Summary: Building a time series model

- Start with obvious seasonal/trend components and choose an additive or multiplicative model. (Plot your data!)
- Examine the usual diagnostic plots, and plot your residuals as a function of time. Do you need a (different) nonlinear time trend? A transformation of Y ?
- Check your residuals for autocorrelation. If it's present, add appropriate lag terms to your model.

Basic time series concepts

Additive decomposition models

Multiplicative decomposition models

A warning about trend-based forecasting

Autoregressive models

Special kinds of $AR(1)$ models

Making predictions with AR models

Combining decomposition and AR models

The market model



The market model

- One final model is of interest: the **market model** for a stock estimates its volatility/risk relative to the market as a whole
- Important case of regressing **one time series on another**
- Predict the stock's percentage change (return) as a function of the percentage change of a market index (e.g., DJIA, S&P 500)
- The slope coefficient is the “beta” of the stock:
 - $\beta > 1$ means the stock is riskier/more volatile than the market as a whole
 - $\beta = 1$ means the stock is as risky/as volatile as the market as a whole
 - $\beta < 1$ means the stock is less risky/less volatile than the market as a whole



Market model for Dell stock

First let's calculate the percentage returns:

```
dell <- dell %>%  
  mutate(dell.return=(dell-lag(dell))/lag(dell),  
         SP500.return=(SP500-lag(SP500))/lag(SP500))
```

Now let's predict Dell's returns from the S&P 500:

```
dell.model <- lm(dell.return ~ SP500.return, data=dell)
summary(dell.model)
```

Call:

```
lm(formula = dell.return ~ SP500.return, data = dell)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.05157	-0.00663	-0.00134	0.00912	0.03577

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.00217	0.00183	-1.18	0.24
SP500.return	1.09719	0.18946	5.79	0.00000026 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

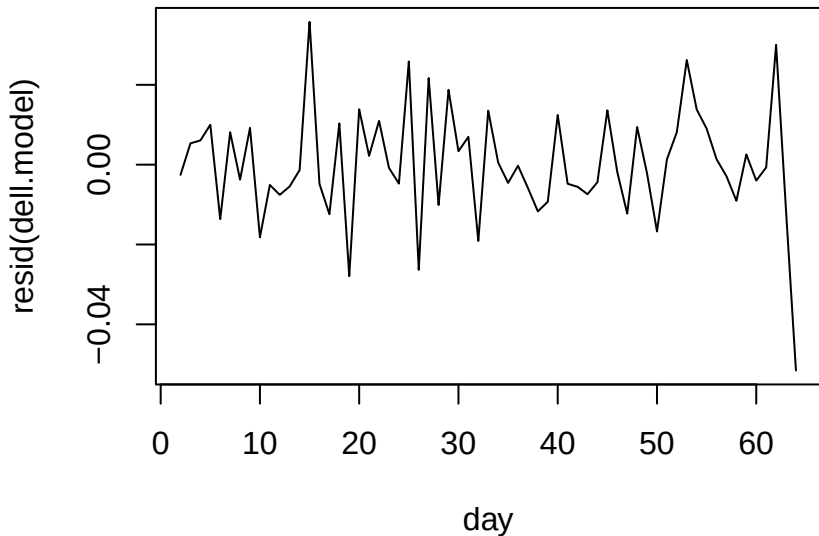
Residual standard error: 0.0144 on 61 degrees of freedom
(1 observation deleted due to missingness)

Multiple R-squared: 0.355, Adjusted R-squared: 0.344

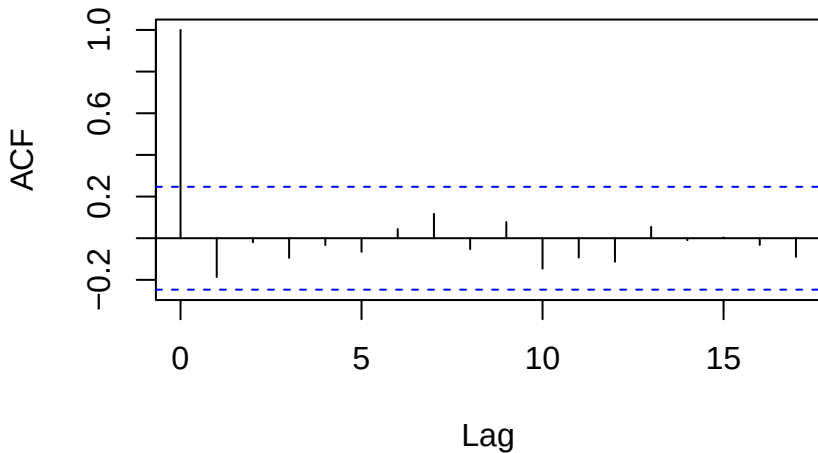
F-statistic: 33.5 on 1 and 61 DF, p-value: 0.000000262

Standard diagnostics look pretty good. How about time dependence in residuals?

```
plot(resid(dell.model)~day, data=dell[-1,], type='l')
```



```
acf(resid(dell.model))
```



These look good!



- Dell's β of 1.10 indicates it's about as volatile/risky as the market as a whole
- The R^2 of 0.36 indicates that about 36% of the variability in Dell's stock price can be attributed to the market's movement
- Since $1 - R^2 = 0.64$, about 64% of the variability in Dell's stock price is from Dell-specific (or industry-specific factors)